

Day 100 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Indian Economy & TN Dev. Admin. (8): UPI's 2026 Records & Global Dominance

READING MATERIAL

Covers UPI's record-breaking 2026 transaction volumes and its position as the world's largest real-time payments platform.

Q: What record did UPI set in May 2026?

A: ₹29.90 trillion in transaction value — the highest monthly transaction value since UPI's 2016 launch — alongside 23.2 billion transactions (a 4% increase over the previous month).

Q: What were UPI's full-year 2025 statistics?

A: 228.5 billion transactions (a 33% year-on-year increase) worth a total of ₹299.74 trillion.

Q: What is UPI's standing among the world's real-time payment systems?

A: The world's **LARGEST** real-time payments platform — accounting for nearly half of all real-time transactions worldwide.

Q: What share of India's total digital payment volume does UPI command, and how many unique users does it have (early 2026)?

A: Nearly 85% of India's total digital payment volume; over 500 million unique users.

Q: What is UPI's projected daily transaction volume by FY2028?

A: 1 billion transactions per day.

MOCK TEST (WITH ANSWERS)

1. UPI's record monthly transaction value, set in May 2026, was approximately:

- A. ₹19.90 trillion
- B. ₹24.90 trillion
- C. ₹29.90 trillion ✓**
- D. ₹34.90 trillion

Explanation: ₹29.90 trillion. [hard, web-research]

2. In full-year 2025, UPI processed how many total transactions?

- A. 128.5 billion
- B. 178.5 billion
- C. 228.5 billion ✓**
- D. 278.5 billion

Explanation: 228.5 billion (33% YoY growth). [hard, web-research]

3. UPI is recognized as the world's largest real-time payments platform, accounting for approximately what share of global real-time transactions?

- A. Nearly 25%
- B. Nearly 50% ✓**
- C. Nearly 75%
- D. Nearly 90%

Explanation: Nearly half (50%). [medium, web-research]

4. UPI commands approximately what share of India's total digital payment volume?

- A. 55%
- B. 65%
- C. 75%
- D. 85% ✓**

Explanation: Nearly 85%. [medium, web-research]

5. UPI is projected to reach 1 billion transactions per DAY by:

A. FY2026

B. FY2027

C. FY2028 ✓

D. FY2030

Explanation: FY2028. [hard, web-research]

Part B – Aptitude & Mental Ability: Lowest Common Multiple (LCM)

READING MATERIAL

LCM is 1 of 10 named Aptitude sub-skills. Real exam evidence shows two distinct shapes: algebraic LCM-from-a-clue problems, and 'remainder' word problems — both are taught below as separate methods.

Q: Real exam (2025): If the LCM of 'a' and 'a+2' is 1300, find the LCM of 'a' and 'a+1'.

A: Since a and a+2 differ by 2, their HCF is either 1 or 2. If $\text{HCF}=2$: $\text{LCM} = a(a+2)/2 = 1300 \rightarrow a(a+2) = 2600$. Solving $a^2+2a-2600=0$ gives $a=50$ (check: $50 \times 52/2 = 1300$ ✓). Now find $\text{LCM}(50, 51)$: consecutive integers are always coprime ($\text{HCF}=1$), so $\text{LCM} = 50 \times 51 = 2550$.

Q: Method — 'least number leaving given remainders' (remainders differ from the divisors by a constant): Find the least number which, when divided by 16, 18, and 21, leaves remainders 3, 5, and 8 respectively.

A: Check the gap between each divisor and its remainder: $16-3=13$, $18-5=13$, $21-8=13$ — all the same gap! When that happens, the answer is $\text{LCM}(16,18,21) - 13$. $\text{LCM}(16,18,21) = 1008$. Answer = $1008-13 = 995$.

Q: Method — 'least number leaving the SAME remainder in every case': Find the least number which, when divided by 6, 7, 8, 9, and 12, leaves the same remainder 1 in every case.

A: When the remainder is identical across all divisors, the answer is $\text{LCM}(\text{divisors}) + \text{remainder}$. $\text{LCM}(6,7,8,9,12) = 504$. Answer = $504 + 1 = 505$.

Q: Method — 'mixed condition' (a common remainder for some divisors, but exact division for another): Find the least number which, when divided by 5, 6, 7, and 8, leaves remainder 3 in each case, but leaves NO remainder when divided by 9.

A: First find the least number leaving remainder 3 for 5,6,7,8: that's $\text{LCM}(5,6,7,8)+3 = 840+3 = 843$. But we also need it divisible by 9. Keep adding $\text{LCM}(5,6,7,8)=840$ repeatedly to 843 until the result is divisible by 9: 843 (no), 1683 ($1683 \div 9 = 187$, yes!). Answer = 1683.

Q: Method — LCM of numbers given as prime factorizations: Find the LCM of $(2 \times 3 \times 5 \times 7)$ and $(3 \times 5 \times 7 \times 11)$.

A: LCM takes the HIGHEST power of every prime that appears in either number. Primes involved: 2,3,5,7,11 — each appears at most once in either factorization, so $\text{LCM} = 2 \times 3 \times 5 \times 7 \times 11$ (just the union of all prime factors, since none repeat).

MOCK TEST (WITH ANSWERS)

1. If the LCM of 'b' and 'b+2' is 180, and their HCF is 2, find the LCM of 'b' and 'b+1'.

A. 342 ✓

B. 380

C. 361

D. 324

Explanation: $\text{LCM} = b(b+2)/\text{HCF} = 180 \rightarrow b(b+2) = 360 \rightarrow b^2+2b-360=0 \rightarrow b=18$ (check: $18 \times 20/2 = 180$ ✓). Now $\text{LCM}(18,19)$: consecutive integers are coprime, so $\text{LCM} = 18 \times 19 = 342$. [hard, authored]

2. Find the least number which, when divided by 12, 15, and 20, leaves remainders 7, 10, and 15 respectively.

A. 55 ✓

B. 60

C. 65

D. 53

Explanation: Gaps: $12-7=5$, $15-10=5$, $20-15=5$ — all equal, so answer = $\text{LCM}(12,15,20) - 5 = 60 - 5 = 55$. [medium, authored]

3. Find the least number which, when divided by 4, 5, and 6, leaves the same remainder 2 in each case.

A. 58

B. 60

C. 62 ✓

D. 64

Explanation: $\text{LCM}(4,5,6) = 60$. Answer = $60 + 2 = 62$. [easy, authored]

4. Find the LCM of $(2 \times 2 \times 3 \times 5)$ and $(2 \times 3 \times 3 \times 7)$.

A. $2 \times 2 \times 3 \times 3 \times 5 \times 7$ ✓

B. $2 \times 3 \times 5 \times 7$

C. $2 \times 2 \times 3 \times 5 \times 7$

D. $4 \times 9 \times 5 \times 7$

Explanation: Take the highest power of each prime: 2^2 (from the first), 3^2 (from the second), 5^1 , 7^1 . $\text{LCM} = 2^2 \times 3^2 \times 5 \times 7$, which equals option (A) written out as $2 \times 2 \times 3 \times 3 \times 5 \times 7$. [medium, authored]

5. The LCM of two consecutive integers n and $n+1$ is always:

A. $n + 1$

B. $n(n+1)$ ✓

C. $n - 1$

D. Cannot be determined

Explanation: Consecutive integers are always coprime ($\text{HCF}=1$), so their LCM is simply their product, $n(n+1)$. [easy, authored]

Part C – General English: Poem: "The Spider and The Fly" — Mary Howitt

READING MATERIAL

A classic cautionary fable-poem: a cunning spider flatters a fly into its web. Real exam evidence (confirmed author attribution via a real 2024 question) tests a wider range of figure-of-speech labels than most other poems in this syllabus — including Apostrophe, a device not seen in earlier topics.

Q: Real exam: Who wrote "The Spider and The Fly"?

A: Mary Howitt (confirmed via a real 2024 'match the poets with their poems' question).

Q: What is the poem's plot and moral?

A: A spider repeatedly flatters and invites a fly into its 'parlour' (web), praising her appearance. The fly initially refuses, recognizing the danger ('who goes up your winding stair can ne'er come down again'), but is eventually won over by enough flattery and is caught.

Moral: beware of flattery from those with bad intentions — once lured in, there's no escape.

Q: What is 'Apostrophe' as a figure of speech, and where does it appear in this poem?

A: Apostrophe is direct address — speaking TO someone (or something personified) as if they were present and listening. "Will you walk into my parlour?" (the spider speaking directly to the fly) and "Oh no, no," said the little Fly... "to ask me is in vain" (the fly's direct reply) are both cited examples — basically, the poem's dialogue-driven structure itself is built on this device.

Q: Identify the figure of speech: "How handsome are your gauzy wings, how brilliant are your eyes!"

A: Personification — the fly's physical features are praised in human terms of 'handsome' beauty, and more broadly the insects throughout the poem speak and reason like humans.

Q: Identify the figure of speech: "Your eyes are like the diamond bright."

A: Simile — directly compares the fly's eyes to a diamond using 'like'.

Q: Identify the figure of speech: "thinking only of her brilliant eyes, and green and purple hue."

A: Imagery — vivid sensory (visual) description of the fly's colours, appealing directly to the reader's visual imagination.

Q: Identify the figure of speech: "For who goes up your winding stair can ne'er come down again."

A: Hyperbole — an absolute, exaggerated claim ('ne'er' = never) used for dramatic warning effect.

MOCK TEST (WITH ANSWERS)

1. "The Spider and The Fly" was written by:

- A. Sarojini Naidu
- B. Mary Howitt ✓**
- C. Emily Dickinson
- D. Jane Taylor

Explanation: Mary Howitt — confirmed via a real 2024 TNPSC question. [easy, question-bank-2024]

2. What is the moral of the poem?

- A. Hard work always pays off
- B. Beware of flattery from those with bad intentions ✓**
- C. Friendship conquers all
- D. Beauty is only skin deep

Explanation: Beware of flattery — the spider lures the fly through repeated compliments. [medium, web-research]

3. "Will you walk into my parlour?" — spoken directly to the fly — is an example of which figure of speech?

- A. Simile
- B. Personification
- C. Apostrophe ✓**
- D. Metaphor

Explanation: Apostrophe — direct address. [hard, web-research]

4. "Your eyes are like the diamond bright" uses which figure of speech?

- A. Simile ✓**
- B. Metaphor

C. Personification

D. Oxymoron

Explanation: Simile — uses 'like'. [easy, web-research]

5. "For who goes up your winding stair can ne'er come down again" is an example of:

A. Irony

B. Alliteration

C. Anaphora

D. Hyperbole ✓

Explanation: Hyperbole — an absolute, exaggerated warning. [medium, web-research]

6. "Thinking only of her brilliant eyes, and green and purple hue" is best described as:

A. Imagery ✓

B. Metaphor

C. Consonance

D. Assonance

Explanation: Imagery — vivid visual/sensory description. [medium, web-research]